A dimension-independent Einstein repair corollary

24 September 2026. This is a short extension of the uniformly local repair theorem, not a new proof of its weighted inverse. That proof is dimension-independent once its local harmonic charts are supplied. The four-dimensional topological mechanism for producing those charts is not used here.

1. Uniformly local repair in dimension $n \geqslant 3$

Fix $n \geqslant 3$, $0 < \alpha < 1$, $\rho > 0$, $\Lambda, C_G, \lambda > 0$, and $d, H \geqslant 0$, with

$$H^2 < 8\lambda. \tag{1}$$

A uniformly controlled harmonic atlas means harmonic coordinate charts centred at every point, with nested coordinate balls of radii ρ and 2ρ, uniform ellipticity $\Lambda^{-1} I \leqslant (g_{ij}) \leqslant \Lambda I$, and coefficient $C^{2,\alpha}$ bounds at most Λ on the larger balls. Uniformly local norms are suprema over the smaller charts, not sums. The constants below depend on these atlas data.

Corollary 1. There exist $\varepsilon, C > 0$, depending only on the fixed data above, with the following property. Let (M^n, g) be a closed connected smooth Riemannian manifold with such an atlas and

$$\operatorname{Vol}_g B_R(x) \leqslant C_G (1+R)^d e^{HR} \quad \text{for all } x \in M,\ R \geqslant 0. \tag{2}$$

For the full Bianchi-gauged linearization

$$L_g = \tfrac{1}{2}\Delta_{L,g} + I, \qquad \Phi_g(h) = \operatorname{Ric}_h + h + \delta_h^*(B_g h), \quad B_g u = \delta_g u + \tfrac{1}{2} d \operatorname{tr}_g u,$$

assume positive quadratic coercivity on **all** symmetric two-tensors:

$$Q_g(u) := \int_M \langle L_g u, u \rangle \, d\mu_g \geqslant \lambda \|u\|^2_{L^2(g)}. \tag{3}$$

If $E = \operatorname{Ric}_g + g$ satisfies

$$\|E\|_{C^{0,\alpha}_{\mathrm{uloc}}} < \varepsilon,$$

then there is a smooth metric h on the same smooth manifold with

$$\operatorname{Ric}_h = -h, \qquad \|h - g\|_{C^{2,\alpha}_{\mathrm{uloc}}} \leqslant C \|E\|_{C^{0,\alpha}_{\mathrm{uloc}}}. \tag{4}$$

Neither orientation, π_2, gauge-theoretic classes, nor a diameter, total-volume or total-L^2-residual bound is assumed.

Proof. The Lichnerowicz operator has principal part $\frac{1}{2}\nabla^*\nabla$ in every dimension. Choose $H/2 < a < \sqrt{2\lambda}$ and put $\phi_x = e^{-a \operatorname{dist}_g(x,\cdot)}$. The weighted identity

$$Q_g(\phi_x u) = \int_M \phi_x^2 \langle L_g u, u \rangle + \tfrac{1}{2} \int_M |d\phi_x|^2 |u|^2$$

and (2)–(3) give, exactly as in the preceding theorem,

$$\|u\|_{L^2(B_1(x))} \leqslant \frac{e^a \sqrt{J_a}}{\lambda - a^2/2} \|L_g u\|_\infty, \qquad J_a = C_G e^H \sum_{j \geqslant 0} (j+2)^d e^{-(2a-H)j} < \infty. \tag{5}$$

Local Moser and Schauder estimates in the supplied atlas therefore yield

$$\|L_g^{-1} f\|_{C^{2,\alpha}_{\mathrm{uloc}}} \leqslant C_I \|f\|_{C^{0,\alpha}_{\mathrm{uloc}}}.$$

Fredholm theory and (3) give this inverse on each closed M. Only the local constants change with n; no dimension enters the absorption coefficient $\lambda - a^2/2$ or the growth condition (1). The uniformly local Hölder quadratic estimate for $\Phi_g(g+u) - E - L_g u$ gives the same contraction proof. Smallness makes $h = g + u$ positive definite and Ric_h negative definite. Finally, $b = B_g h$ satisfies $\nabla_h^* \nabla_h b - \mathrm{Ric}_h(b) = 0$; integration forces $b = 0$. Elliptic bootstrapping gives smoothness. □

If additionally $\|\nabla \mathrm{Ric}_g\|_\infty \leqslant K_1$, the atlas gives $\|E\|_{C^{0,\alpha}_{\mathrm{uloc}}} \leqslant C\delta_-^{1-\alpha}$ for $\delta_- \leqslant 1$. Thus small pointwise Ricci defect alone suffices with these same fixed geometric and operator data. An absolute spectral gap is not (3), and genuine kernels have not been removed.

2. Actual large-diameter examples in every dimension $n \geqslant 3$

Fix a closed hyperbolic (N^n, h), normalized by $\mathrm{Ric}_h = -h$, and an epimorphism $\pi_1(N) \to \mathbb{Z}$. Millson's compact examples provide such a choice in every $n \geqslant 3$. Let (N_m, h_m) be the finite cyclic covers. The fixed infinite cyclic cover has a proper cocompact $\mathbb{Z}$-action, so counting its bounded fundamental regions gives

$$\mathrm{Vol}_{h_m} B_R(x) \leqslant C_N(1+R), \qquad \mathrm{Vol}(N_m, h_m) = m\,\mathrm{Vol}(N, h).$$

Consequently their diameters tend to infinity. This bound concerns the actual cyclic quotients, not their universal covers or arbitrary covers.

For sectional curvature $-1/(n-1)$, the classical Koiso estimate gives

$$Q_{h_m}(u) \geqslant \lambda_n \|u\|_2^2, \qquad \lambda_n = \frac{n-2}{2(n-1)} > 0. \tag{6}$$

This is Jäckel's Lemma 2.4, rescaled from $\mathrm{Ric} = -(n-1)g$; its quadratic estimate, unlike his principal theorem's stated dimensional scope, also applies in dimension three.

All sufficiently small $C^3_{\mathrm{uloc}}(h_m)$ perturbations g_m have uniform harmonic atlas and Ricci-derivative bounds, satisfy (2) with $H = 0, d = 1$, and satisfy (3) with $\lambda = \lambda_n/2$. The last assertion follows from the almost-Einstein Koiso estimate and uniform negative sectional curvature; it is not spectral continuity with a constant depending on m. Corollary 1 supplies a uniform repair threshold independent of m.

The residual can genuinely be distributed over all cells. For disjoint lifts β_j of a fixed bump, equal to one on a smaller ball, take $f_m = t\sum_{j=1}^m c_j \beta_j$, $1 \leqslant c_j \leqslant 2$, and $g_m = e^{2f_m} h_m$. On the smaller balls,

$$|\mathrm{Ric}_{g_m} + g_m|_{g_m}^2 \, d\mu_{g_m} = n(e^{2tc_j} - 1)^2 e^{(n-4)tc_j} \, d\mu_{h_m}.$$

Hence $\|E_m\|_2 \geqslant ct\sqrt{m}$ while $\|E_m\|_{C^{0,\alpha}_{\mathrm{uloc}}} \leqslant Ct$, uniformly in m. Small fixed t is allowed despite the divergent total residual.

This corollary makes one transferable analytic module explicit. It gives uniform correction on known hyperbolic smooth types and a general conditional solver, not new Einstein topologies or a dimension-only Yau26 threshold. In higher dimensions, the controlled atlas remains an input; the four-dimensional topological regularity theorem has not been extended by this argument.